

5C Adding and subtracting terms

Learning Objective	Apply algebraic properties to rearrange, expand and factorise linear expressions
Today We Are	Simplifying expressions by combining like terms
Indicators of Progress	I can add and subtract like terms

15

Magic Sum = 15

8	1	6
3	5	7
4	9	2

18

Magic Sum = 18

10	1	7
3	6	9
5	11	2

Lesson starter: Like terms

- Put these terms into groups of like terms: ~~4a~~ 5b ~~2ab~~ ~~3/a~~ ~~2a~~ 7b² 5a²b 9aba

4a	2ab	5b	7b ²	5a ² b
2a	3ba			9aba

$$6 \times 6$$

$$a \times a = a^2$$

Key Ideas

- Like terms** contain exactly the same pronumerals with the same powers; the pronumerals do not need to be in the same order; for example, 4ab and 7ba are like terms.
- Like terms can be combined when they are added or subtracted to simplify an expression; for example, 3xy + 5xy = 8xy.
- The sign in front of the term stays with the term even when it is moved.

$$3x + 7y - 2x + 3y + x - 4y = 3x - 2x + x + 7y + 3y - 4y$$

$$= 2x + 6y$$



$$= 6t \qquad = 6x + 10y \qquad = 5ac + 2b$$

d) $2x^2y + 10x - 6x^2y + 4xy^2$
 $= -4x^2y + 10x + 4xy^2$



